Team25 I2S Driver

IO

Name	Size	Input/Output	Function
m_clk	1	input	Main clock (12MHz from FPGA)
s_clk	1	output	Serial clock (1.5MHz)
lr_clk	1	output	Left/right select clock (46.875kHz)
rst	1	input	Reset
sda_rx	1	input	Serial data in
sda_tx	1	output	Serial data out
rsamp_rx	16	input	16-bit audio sample for right channel
rsamp_tx	16	output	16-bit audio sample for right channel
lsamp_rx	16	input	16-bit audio sample for left channel
lsamp_tx	16	output	16-bit audio sample for left channel

Clock Generation

The frequency of the main clock m_clk will reflect the clock speed of the Cmod A7 FPGA, which is 12MHz. The other two clock inputs s_clk and lr_clk will be generated by clock dividing modules that will slow m_clk by 8 times and 256 times respectively.

Deserializer (Rx)

The driver receiving serial data from the Pmod ADC will deserialize the data by shifting serial data into two 16-bit shift registers, one for the left channel and one for the right channel. Separating the left and right channel data into two shift registers allows for the other channel to immediately shift in new data while simultaneously allowing the sample that was just collected to be added to the 48-sample buffer that becomes the multiplicand in the convolution operation. If

lr_clk is high, s_data gets shifted into r_sample, and if lr_clk is low, s_data gets shifted into l_sample instead. The module controlling the 48-bit buffer can see whether a channel has a valid sample based on the state of lr_clk.

Serializer (Tx)

The driver transmitting serial data to the Pmod DAC will serialize the data by shifting data out of two 16-bit shift registers. See the description of the deserializer for the rationale of having separate shift registers for the left and right channels.